HACKROME DEMO SOURCE BRIEF

Netify

The AI layer that turns PDFs into narrated, animated video lessons.

ELEVATOR PITCH

Turn any learning PDF into a custom animated teacher-led video.

DEMO PROOF

The submitted explainer video should be generated by Netify itself.

HACKATHON WORK

ElevenLabs narration, Supabase product layer, web app flow, and API foundation.

CURRENT STATUS

Local generation works; durable cloud rendering is the next step.

This PDF is written so Netify can generate a Devpost-ready demo video from it. It emphasizes the personal learning origin, the teacher voice, the HackRome build, and the honest self-generated proof.

1. The New Story Angle

Netify started from a personal learning problem: I have high school exams in a week, and I learn better from videos than from static documents.

A PDF can contain everything a student needs to know, but a PDF does not always teach. A good lesson has pacing, emphasis, examples, visuals, and a voice that guides the learner through the material. Netify exists to give documents that teaching layer.

The broader product insight is that students, edtech startups, and corporate training teams all face the same bottleneck. They already have written learning content, but producing useful video lessons is slow, expensive, and hard to scale.

> **Main perspective:** Netify is not just PDF-to-video. Netify is PDF-to-video lessons. The goal is teaching, not conversion.

2. What Netify Does

Netify transforms documents into narrated video lessons. A user uploads a PDF. Netify parses the document, extracts the core ideas, plans how to teach them, generates custom animated scenes, creates narration with ElevenLabs, renders the final MP4, and streams progress while the job runs.

The output is not a slide deck with a voice track. Each scene is generated around the function it needs to serve: explaining a concept, comparing two ideas, showing a process, highlighting a statistic, or making dense material easier to remember.

For students

Turn exam PDFs, lecture notes, and dense study material into guided video lessons.

For edtech

Add video lesson generation without building a media pipeline from scratch.

For training

Convert onboarding, compliance, and internal knowledge into watchable instruction.

3. Why It Matters

Edtech is full of written content. Financial education PDFs, language-learning notes, corporate onboarding guides, compliance policies, exam summaries, and course handouts all share the same weakness: they require the learner to create structure and momentum alone.

Video lessons solve that problem by adding guidance. They decide what comes first, what matters most, what example makes the concept click, and when the learner needs visual reinforcement. Netify automates that transformation.

Static document

- Information exists, but pacing is missing.
- The learner must decide what matters.
- Dense sections are easy to skip.
- No teacher voice or emotional guidance.

Netify lesson

- Ideas are sequenced into scenes.
- Visuals are matched to the teaching job.
- ElevenLabs narration gives the lesson a voice.
- The final result is an MP4 people can watch.

4. The Product Built During HackRome

Netify existed before HackRome as a local generation pipeline. During the hackathon, the focus was turning that pipeline into the beginning of a real product.

Area	HackRome progress
Teacher narration	Integrated ElevenLabs TTS with `eleven_multilingual_v2`, scene-level MP3 generation, retry logic, and audio-duration detection.
Product backbone	Connected Supabase Auth, Postgres tables, private resource storage, private video storage, and row-level security.
Web app flow	Built a local flow where users can sign in, upload PDFs, store resources, start generation, watch progress, and view generated videos.
Developer API	Added the foundation for asynchronous render jobs, Bearer auth, signed PDF handoff, SSE progress, and Supabase upload of completed MP4s.
Render reliability	Improved validation and repair loops for generated Remotion/TypeScript scenes so the product can recover from common generation failures.

5. How the Pipeline Works

1. **Upload a learning PDF**
   The source can be study material, a course PDF, financial literacy content, onboarding material, or a corporate training document.

2. **Understand the material**
   LLM agents parse the document, identify the core ideas, and decide what a learner needs to understand.

3. **Plan how to teach it**
   The system creates a lesson structure, not only a summary. It decides what deserves a scene, what example helps, and what visual form fits.

4. **Generate custom animated scenes**
   Netify writes Remotion/TypeScript scene code with layouts, motion, icons, charts, comparisons, and transitions based on the source material.

5. **Add ElevenLabs teacher narration**
   Each scene gets an MP3 voiceover. The narration makes the video feel guided instead of silent or mechanical.

6. **Render, repair, and deliver**
   Remotion renders the final MP4. If generated code fails, the repair layer diagnoses the scene or transition, patches it, and retries.

6. Why ElevenLabs Is Central

Voice is not a small add-on for an education product. A lesson needs a teacher voice. ElevenLabs changes the emotional quality of the output because the final video feels guided, not just animated.

For the demo, emphasize that ElevenLabs is what turns Netify from "moving text" into instruction. The voice gives pacing, confidence, and a human sense of direction.

> **Prize positioning:** ElevenLabs is the strongest sponsor fit because the hackathon-new TTS integration directly improves Netify's core user experience.

7. Current Product Surfaces

Local web app

Users can sign in, upload source PDFs into private Supabase storage, start video generation, watch live progress, and return to a generated video library.

Next.js | Supabase Auth | Private storage | Video library

API foundation

A backend can submit a signed PDF URL, start an asynchronous render job, authenticate with a Bearer token, and stream progress with Server-Sent Events.

POST /api/render | GET /api/events/:jobId | Bearer auth | MP4 output

8. Honest Demo Status

The render service currently runs locally. The cloud render backend is designed to become a durable containerized service, but deployment was blocked during the hackathon because the Scaleway setup required a credit card even with hackathon credits available.

This should be framed honestly in the video if needed: Netify is a working local product demo with cloud-ready architecture. The next step is durable hosted rendering so judges, students, and edtech teams can generate videos directly.

> **Strongest proof:** The demo video explaining Netify should itself be generated by Netify from this PDF.

9. Suggested Devpost Video Narrative

The video should feel personal first, product second, infrastructure third. Lead with the exam/student pain, then expand into edtech.

Scene	Main message	Visual direction
1. Personal hook	I have exams soon, and I learn better from videos than static PDFs.	Student desk, dense PDF, exam countdown, video lesson appearing as relief.
2. Bigger problem	Edtech and training teams have content, but producing video is slow.	Stacks of documents on one side, empty video player on the other.
3. Netify promise	Netify turns learning PDFs into narrated animated video lessons.	PDF enters a pipeline and exits as an MP4 with waveform and animated scenes.
4. Teaching pipeline	Understand, plan, generate scenes, narrate, render.	Six-step animated flow with teaching-focused labels.
5. ElevenLabs moment	A video lesson needs a teacher voice.	Audio waveform, voice avatar, scene timing syncing with narration.
6. HackRome build	The local pipeline became a product foundation.	Four cards: ElevenLabs, Supabase, web app, API.
7. Self-demo proof	This video was generated by Netify itself.	Recursive proof: this PDF turning into the video being watched.
8. Close	Content in, video lesson out.	Final generated lesson inside a learning platform.

10. Recommended 90-Second Voiceover

Netify is the AI layer that turns PDFs into video lessons.

I started from a personal problem: I have high school exams in a week, and I learn much better from videos than from static documents. But the same problem exists across edtech. Teams already have learning content, but producing video takes weeks or months.

Netify takes a PDF, understands the material, plans how to teach it, generates custom animated scenes, creates narration with ElevenLabs, renders an MP4, and streams progress while the job runs.

This is not a slide generator. The visuals are generated around the function of each scene: explaining, comparing, showing a process, or making a concept easier to remember.

During HackRome, I turned the local pipeline into the start of a product: ElevenLabs narration, Supabase auth and storage, a web app flow, and the foundation of the Netify API.

The video you are watching was generated by Netify itself. Content in, video lesson out.

11. Phrases to Preserve

- Netify is the AI layer that turns PDFs into narrated, animated video lessons.
- Turn any learning PDF into a custom animated teacher-led video.
- A PDF can contain everything, but it does not always teach.
- This is not a slide generator.
- A lesson needs a teacher voice.
- During HackRome, the local pipeline became the start of a real product.
- The demo video was generated by Netify itself.
- Content in, video lesson out.

12. What to Avoid in the Video

- Do not make the story only about API infrastructure. The Devpost angle is learning, teaching, and the HackRome build.
- Do not overclaim that the hosted cloud service is fully deployed. The honest current state is local render with cloud-ready architecture.
- Do not say Netify uses OpenAI models unless the submitted demo actually does. The current backend uses DeepSeek through an OpenAI-compatible interface.
- Do not frame the output as generic AI video. The specific promise is video lessons from learning PDFs.

13. Final Positioning

Netify takes learning content that already exists and turns it into a guided video lesson. It combines document understanding, lesson planning, generated animated scenes, ElevenLabs narration, Remotion rendering, repair loops, Supabase-backed product infrastructure, and an API foundation.

The clearest Devpost proof is simple: use Netify to generate the video that explains Netify.

Prepared from the Netify Devpost submission, API documentation, and current repository state. Intended as source material for a HackRome demo video generated by Netify.